

# Dual-Domain Analysis of Adversarial Sensitivity for Forgery Localization

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## Abstract

*Detecting and localizing manipulated regions in AI-generated images remains challenging, as modern detectors often achieve high real-vs.-fake accuracy while offering limited insight into where their decisions originate, especially under adversarial perturbations. In this work, we introduce a framework for forgery localization based on adversarial vulnerability maps, defined as the pixelwise difference between explainability maps computed on clean images and their adversarial counterparts. The underlying idea is that adversarial attacks expose spatial instabilities in regions that are critical to a detector's decision.*

*We evaluate this approach on the B-Free dataset, considering both same-category and different-category inpainting forgeries with available ground-truth annotations. Vulnerability maps are computed using multiple detectors, attacks, and explainability methods, and their spatial alignment with manipulated regions is assessed through standard localization metrics. Our results show that adversarial vulnerability maps provide localization cues that are consistent with those of standard saliency maps, supporting their use as a complementary tool for explaining model behavior and localizing image forgeries.*

## 1. Introduction

Advances in image synthesis and generative inpainting have dramatically increased the realism of manipulated visual content, making traditional forensic cues increasingly unreliable. Diffusion-based models can alter objects or reconstruct regions with minimal perceptual artifacts, motivating growing interest in AI-generated image detection and forgery localization [2, 5].

Recent studies show that adversarial perturbations can expose fragile or discriminative regions in deep models [14, 15], suggesting that vulnerability—rather than standard

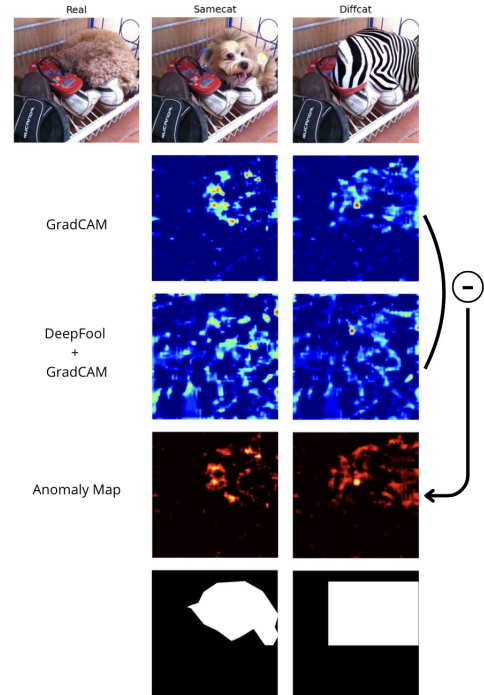


Figure 1. Visualization of the proposed adversarial vulnerability localization pipeline. In this specific visualization, Resnet50 nodown model, GradCAM explainability method and DeepFool adversarial attack are used

explainability—may reveal intrinsic forensic signals.

In this work, we investigate whether the difference between explanations of a clean image and its adversarially perturbed counterpart—termed an adversarial vulnerability map—can serve as an effective cue for manipulation localization. Our hypothesis is that adversarial attacks selectively amplify instability in regions critical for distinguishing real from fake content, causing manipulated areas to exhibit stronger vulnerability than benign regions and poten-

tially yielding better alignment than classical saliency maps.

We conduct our study on the B-Free dataset [5], a carefully curated benchmark designed to minimize semantic and compression biases. B-Free pairs real images with reconstructed and locally manipulated versions, enabling evaluation on visually subtle yet semantically meaningful forgeries.

Using this dataset, we assess multiple forensic detectors [3, 9], adversarial attacks, and explanation methods, measuring ROC, and mass-in-mask overlap to compare vulnerability-based localization against conventional explainability approaches.

## 2. Related Work

**AI-Generated Image Detection and Bias Control.** Recent detectors for AI-generated images often suffer from poor generalization due to dataset bias and spurious correlations. Guillaro et al. [5] address this issue by introducing the B-Free dataset, which pairs real images with self-conditioned diffusion reconstructions and localized inpainting operations, enforcing strong semantic alignment between real and fake samples. This design reduces reliance on superficial cues and provides a controlled benchmark for analyzing detector behavior beyond classification accuracy.

**Adversarial Robustness and Vulnerability.** Adversarial robustness studies show that model sensitivity is spatially heterogeneous. Zhao et al. [14, 15] demonstrate that adversarial attacks expose structured vulnerable regions closely tied to model decisions. In the forensic domain, De Rosa et al. [3] analyze the robustness of CLIP-based detectors, highlighting substantial variation across architectures and attack types, but without addressing spatial localization.

**Frequency-Domain Forensic Cues.** Frequency-based approaches exploit spectral artifacts introduced by generative models, but their explanatory power remains limited. Mandelli et al. [9] show that modern detectors do not primarily rely on isolated spectral peaks, suggesting that discriminative cues emerge from more complex spatial-frequency interactions. Such methods are inherently global and do not provide pixel-level localization.

**Forensic-Oriented Training and Spatial Focus.** Corvi et al. [2] propose forensic-oriented augmentations that encourage detectors to focus on meaningful spatial cues, improving robustness and interpretability. Their work highlights the importance of understanding *where* detectors attend, but does not explicitly exploit adversarial sensitivity for localization.

## 3. Method

### 3.1. Overview

We propose a framework for *forgery localization* that leverages the adversarial sensitivity of AI-generated image detectors. Rather than training a dedicated localization model, we analyze how a detector’s *explainability maps* change under adversarial perturbations. Our core hypothesis is that regions responsible for forged content induce higher instability in the detector’s decision process, and that such instability can be exploited to localize manipulated areas.

Given a detector trained for real-versus-fake classification, we generate adversarial examples, compute explainability maps before and after attack, and define a *vulnerability map* as the pixelwise discrepancy between the two explanations. We then evaluate the spatial alignment between vulnerability maps and ground-truth manipulated regions.

### 3.2. Explainability Maps

Let  $f_\theta$  denote a trained forgery detector. For an input image  $x \in \mathbb{R}^{H \times W \times 3}$ , an explainability method produces a saliency map

$$E(x) \in \mathbb{R}^{H \times W}, \quad (1)$$

which highlights the spatial regions most influential for the detector’s prediction. Depending on the detector architecture, we employ explainability methods such as Grad-CAM, Score-CAM, or other compatible attribution techniques, ensuring that all methods yield maps spatially aligned with the input image.

### 3.3. Adversarial Attacks

To probe detector sensitivity, we generate adversarial examples using standard first-order attacks. Given an image  $x$  and its predicted label  $\hat{y} = f_\theta(x)$ , an adversarial attack  $A$  produces a perturbed image

$$x^{\text{adv}} = A(f_\theta, x), \quad (2)$$

where the perturbation is constrained to be imperceptible under a chosen norm. In our experiments, we consider the following attacks:

- **Fast Gradient Sign Method** [4]: a single-step input perturbation in the direction of the sign of the loss gradient.
- **Projected Gradient Descent** [8]: an iterative application of FGSM.
- **DeepFool** [10]: an iterative estimation of the minimum perturbation necessary to cross the model’s decision boundary.

### 3.4. Adversarial Vulnerability Maps

We define the *adversarial vulnerability map* as the signed difference between the L1-normalized explainability maps  $N(\cdot)$  computed on the clean and adversarial images:

$$V(x) = |N(E(x)) - N(E(x^{\text{adv}}))|. \quad (3)$$

Intuitively,  $V(x)$  highlights regions where the detector’s explanatory focus is most unstable under adversarial perturbation. Our central assumption is that forged regions—introduced through inpainting—are more likely to induce such instability, as they contain generator-specific artifacts or semantic inconsistencies critical to the detector’s decision.

Although we initially considered the absolute difference, we observed that it often introduced noisy high responses, especially in regions where the model had little or no attention in the clean image but increased attention in the adversarial one. For this reason, we use the signed difference, which produces cleaner and more interpretable vulnerability maps.

### 3.5. Forgery Types and Ground Truth

We evaluate our method on localized inpainting forgeries derived from the B-Free dataset [5]. Each real image is paired with two manipulated variants:

- **Same-category inpainting**, where a masked region is replaced with content from the same semantic category, accompanied by a binary mask.
- **Different-category inpainting**, where a region is replaced with an object from a different category, annotated by a bounding box.

All images preserve the original background, ensuring that localization focuses exclusively on the manipulated region. A subset of 800 images for both the same-category and the diff-category sets is considered in our study.

### 3.6. Spatial Correlation Metrics

Let  $S \in \mathbb{R}^{H \times W}$  denote a spatial map (either  $E(x)$  or  $V(x)$ ) and  $M \in \{0, 1\}^{H \times W}$  the ground-truth manipulated region.

$$\begin{aligned} \text{ROC-AUC} &= \text{AUC}(\text{TPR}(\tau), \text{FPR}(\tau)) \\ \text{PR-AUC} &= \text{AUC}(\text{Precision}(\tau), \text{Recall}(\tau)) \\ \text{MF}(S, M) &= \frac{\sum_{i \in M} S_i}{\sum_i S_i} \\ \text{IoU@Top-}k &= \frac{|\text{Top-}k(S) \cap M|}{|\text{Top-}k(S) \cup M|} \end{aligned} \quad (4)$$

ROC-AUC and PR-AUC measure pixel-level discrimination between manipulated and non-manipulated regions by thresholding  $S$ . Mass Fraction quantifies the proportion of saliency mass inside the ground-truth region. Top- $k$  IoU evaluates the spatial overlap between the  $k$  highest-scoring pixels and the manipulated area.

All metrics are computed identically for explainability and adversarial vulnerability maps.

### 3.7. Waverep and LIME

WaveRep [2] is a ViT-based deepfake detector that primarily exploits low-level forensic cues in the frequency do-

main rather than semantic image content. Owing to this design, it generalizes well to previously unseen datasets and achieves strong performance without fine-tuning on the B-free dataset.

However, the frequency-oriented nature of WaveRep limits the effectiveness of gradient-based explanation methods. They indeed produce sparse attribution maps exhibiting weak spatial alignment with the manipulated regions, reflecting the poor correspondence between pixel-space gradients and frequency-domain decision cues.

In contrast, perturbation-based explanation methods are better suited to this setting. We therefore adopt LIME [11] method to SLIC superpixels [1]. For adversarially perturbed images, the same region partitioning is reused to ensure spatial consistency with the clean explanation.

### 3.8. ResNET 50 and GradCAM

With the goal of assessing the vulnerability of different model categories to adversarial attacks, we implement ResNet-50 [6] as a representative CNN-based architecture.

The convolutional structure of the architecture induces a strong bias toward local spatial feature extraction, making it particularly relevant to our setting, which focuses on locally inpainted regions and the spatial structure of the associated adversarial perturbations.

The interpretation of the model’s decisions is performed using Gradient-weighted Class Activation Mapping (Grad-CAM) [12]. In this study, Grad-CAM maps are used to analyze the spatial response of the model under adversarial perturbations and to support the evaluation of vulnerability patterns. This enables a direct comparison between regions that contribute to the model’s decision and those most affected by adversarial attacks, providing a CNN-specific reference for correlation analysis across different model families.

### 3.9. AnomalyOV

As another method to detect AI content generation and extract explainability maps, we test AnomalyOV. This is a model seeking optimal performances in zero-shot anomaly detection, capable of providing meaningful significance maps and visual features enriched by the information about “where the anomalies are” in the image.

Like the authors of the Anomaly OV paper, we believe that the anomaly expert component can be good for any visual task that require spatial image inspection. Nonetheless anomaly detection and forgery detection tasks have a similar intuitive meaning if we consider forgery detection a matter of “seeing visual anomalous clues that tell where and whether a forgery is in the image”

We want to exploit the model’s architecture’s ability to extract significance map within a single forward call, without additional gradient computations or multiple steps

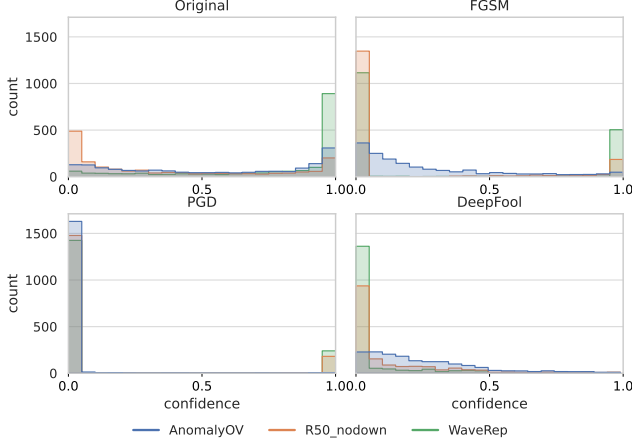


Figure 2. Distribution of sigmoid-normalized logit scores for in-painted images, across detectors under clean conditions and adversarial attacks, illustrating the effect of perturbations on detector decision score.

methods needed [13]. Additionally, its SigLIP encoder splits the image into multiple crops using AnyRes strategy [7]. This allows to independently compute saliency maps across all image’s crops and aggregate them to predict the anomaly score [13]. (Fig. 5)

However since it tends to focus on the semantical subject of the image (which is not necessarily matching the inpainting mask) (Fig. 6), we decided to finetune the anomaly expert’s weights only on a small subset of the diffcat training set ( $\sim 4k$  real-inpainted pairs images) for binary classification task between inpainted (label 1) or real (label 0).

After the finetuning, we integrated AnomalyOV in the overall pipeline using its prediction score and its significance map as explanation map.

## 4. Results

### 4.1. Decison Score Distributions under Adversarial Attacks

Figure 2 reports the distribution of detector decision scores—of images being classified as fake—under clean conditions (*Original*) and in the presence of adversarial perturbations. Unless otherwise stated, detector outputs are analyzed in terms of their raw logits; for visualization purposes, logits are mapped to the  $[0, 1]$  range through a sigmoid function.

In the clean setting, all detectors exhibit a similar score distribution, characterized by a pronounced peak at high score values and a trailing tail toward lower scores. Adversarial attacks are effective across all detectors, consistently shifting score mass toward lower score regions.

In particular, WaveRep and R50-nodown show nearly identical behavior under attack and retain a non-negligible high-score peak under FGSM and PGD, indicating partial

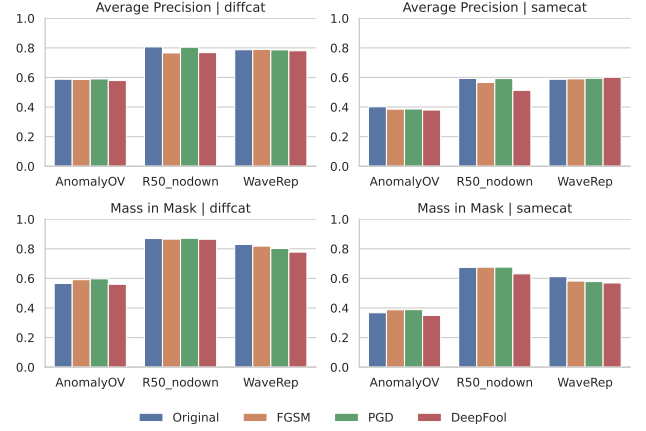


Figure 3. Comparison of localization performance obtained using clean explainability maps (*Original*) and adversarial vulnerability maps induced by FGSM, PGD, and DeepFool. Results are reported in terms of Average Precision and Mass-in-Mask for different-category and same-category inpainting.

robustness adversarial perturbations.

### 4.2. Localization Performance of Explainability and Vulnerability Maps

We report localization results obtained using *adversarial vulnerability maps* in Figure 3. Importantly, all metrics reported in this section refer to vulnerability maps induced by the corresponding adversarial attacks (FGSM, PGD, DeepFool), and *not* to explainability maps computed on the attacked images themselves. The *Original* setting refers exclusively to explainability maps extracted from clean, non-adversarial inputs.

Across detectors and attack types, vulnerability-based localization exhibits a clear and consistent structure that closely mirrors the behavior observed for localization using clean explainability maps.

**Evaluation Protocol and Metrics.** Localization performance is assessed using Average Precision (AP) and Mass-in-Mask (MiM), which show highly correlated behavior and are therefore discussed jointly. To focus the analysis on samples for which localization is meaningful, we restrict evaluation to the top 50% highest-confidence predictions on clean images for each detector and manipulation category. This choice reflects the strong dependency of localization quality on the detector’s ability to correctly identify manipulated content.

For all detectors, localization performance is consistently higher for *different-category* inpainting than for *same-category* manipulations. This effect is partly explained by the larger spatial extent and lower structural complexity of the rectangular bounding boxes used in

different-category forgeries, compared to the finer-grained segmentation masks characterizing same-category inpainting.

**Effect of Detector Architecture.** WaveRep and R50-nodown consistently achieve the strongest localization performance across both AP and MiM. Their maps reveal structured, spatially coherent regions that correlate with forged areas and show minimal sensitivity to the specific adversarial attack used to induce them.

In contrast, AnomalyOV exhibits limited localization capability in both explainability and vulnerability settings. Although fine-tuning improves classification behavior, its significance maps tend to focus on dominant semantic subjects rather than localized inpainting regions, resulting in weak spatial correlation with ground truth. This suggests that anomaly-oriented saliency is not well suited for either explainability or vulnerability-based forgery localization.

**Consistency between Clean and Vulnerability-Based Localization.** A key observation is that vulnerability-based localization performance is fully consistent with the behavior observed using clean explainability maps. Detectors that exhibit strong localization in the clean setting also generate informative vulnerability maps, while detectors with poor clean localization do not benefit from adversarial sensitivity analysis. Adversarial perturbations therefore act as a *contrast-enhancing mechanism*, amplifying pre-existing spatial sensitivities of the detector rather than introducing novel or spurious localization cues.

Differences among FGSM, PGD, and DeepFool remain secondary compared to inter-detector differences, reinforcing the conclusion that localization performance is driven primarily by detector architecture rather than by the specific attack dynamics and explainability method.

## 5. Limitations

The proposed analysis is subject to several limitations. First, the experimental evaluation is conducted on a restricted set of detector architectures and explainability methods. While this choice allows for controlled comparisons, it limits the ability to draw fully general conclusions about the behavior of adversarial vulnerability maps across different model families.

Moreover, the use of a fine-tuned anomaly expert within AnomalyOV introduces a potential bias inherited from its original anomaly detection objective, which may favor semantic saliency over localized forgery cues. This affects the interpretability of the resulting significance maps in the context of inpainting localization.

Finally, both perturbation-based explanations and vulnerability analysis are performed in the image domain with-

out explicitly controlling the contribution of frequency-domain components, despite their known relevance in forensic detection.

## 6. Future Work

Future work will extend the proposed framework by evaluating a wider range of detector and explainability method combinations, in order to assess the generality of adversarial vulnerability maps across architectures. We also plan to explore alternative training strategies for anomaly-based detectors, including training the anomaly expert from scratch with objectives explicitly tailored to forgery localization.

In addition, adversarial perturbations will be directly integrated into perturbation-based explainability methods, replacing standard LIME noise with attack-driven perturbations that better reflect the detector’s decision boundaries.

We also believe that we should explore more methods to extract vulnerability maps using adversarial attacks, even though doing so could completely change our analysis pipeline (for instance, explainability maps could no longer be needed).

Finally, we aim to study the influence of frequency-domain manipulations on vulnerability map generation to better understand the interaction between spatial and spectral forensic cues.

## 7. Conclusions

Adversarial vulnerability maps provide stable and meaningful localization cues that closely follow the intrinsic spatial sensitivity of the detector. Localization performance under adversarial vulnerability analysis remains coherent with clean explainability-based localization and consistent across attack methods. These results support adversarial vulnerability as a reliable signal for forgery localization and as a complementary tool to state-of-the-art explainability methods.

The effectiveness of adversarial vulnerability maps is fundamentally constrained by detector architecture and explainability method couples, indicating that vulnerability amplifies existing spatial inductive biases rather than compensating for their absence.

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# Dual-Domain Analysis of Adversarial Sensitivity for Forgery Localization

## Supplementary Material

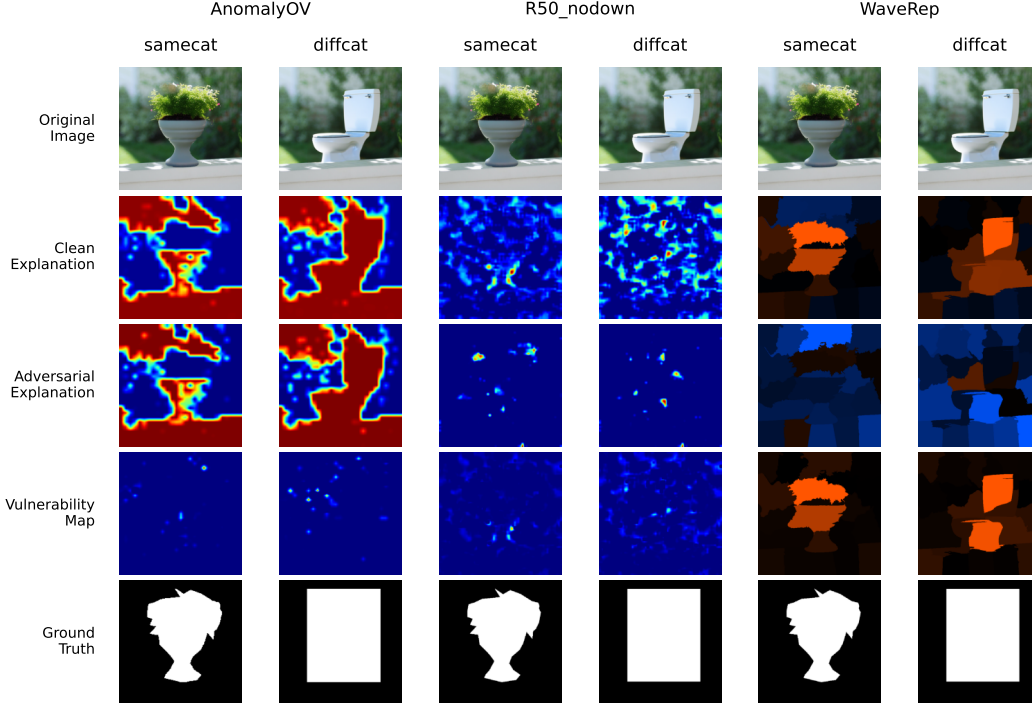


Figure 4. Comparison of the explanation and vulnerability maps of the three detectors, under PGD attack.

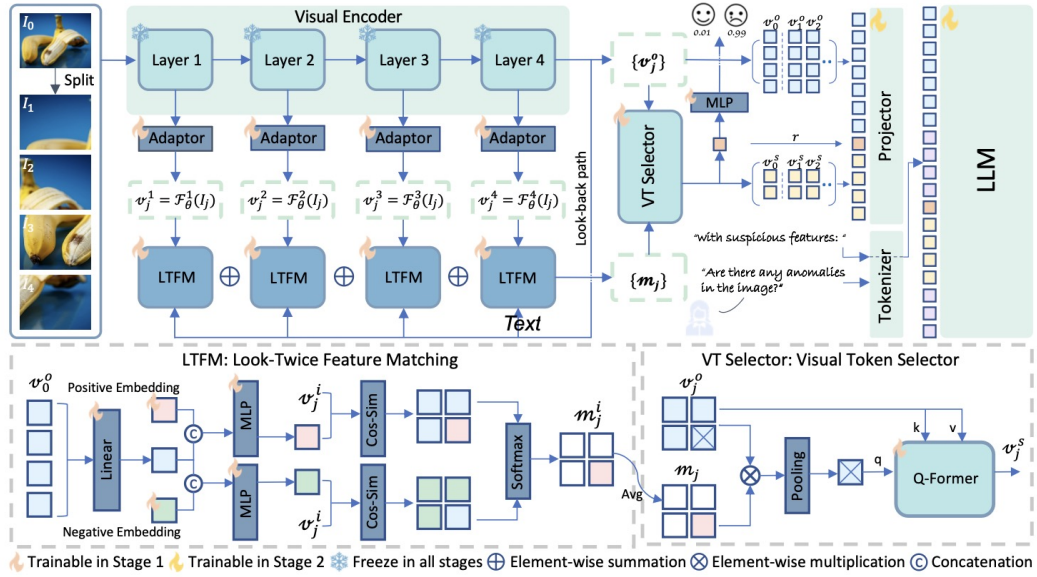


Figure 5. Anomaly OV architecture, we will use only the anomaly expert, thus all the components except the projector and the LLM

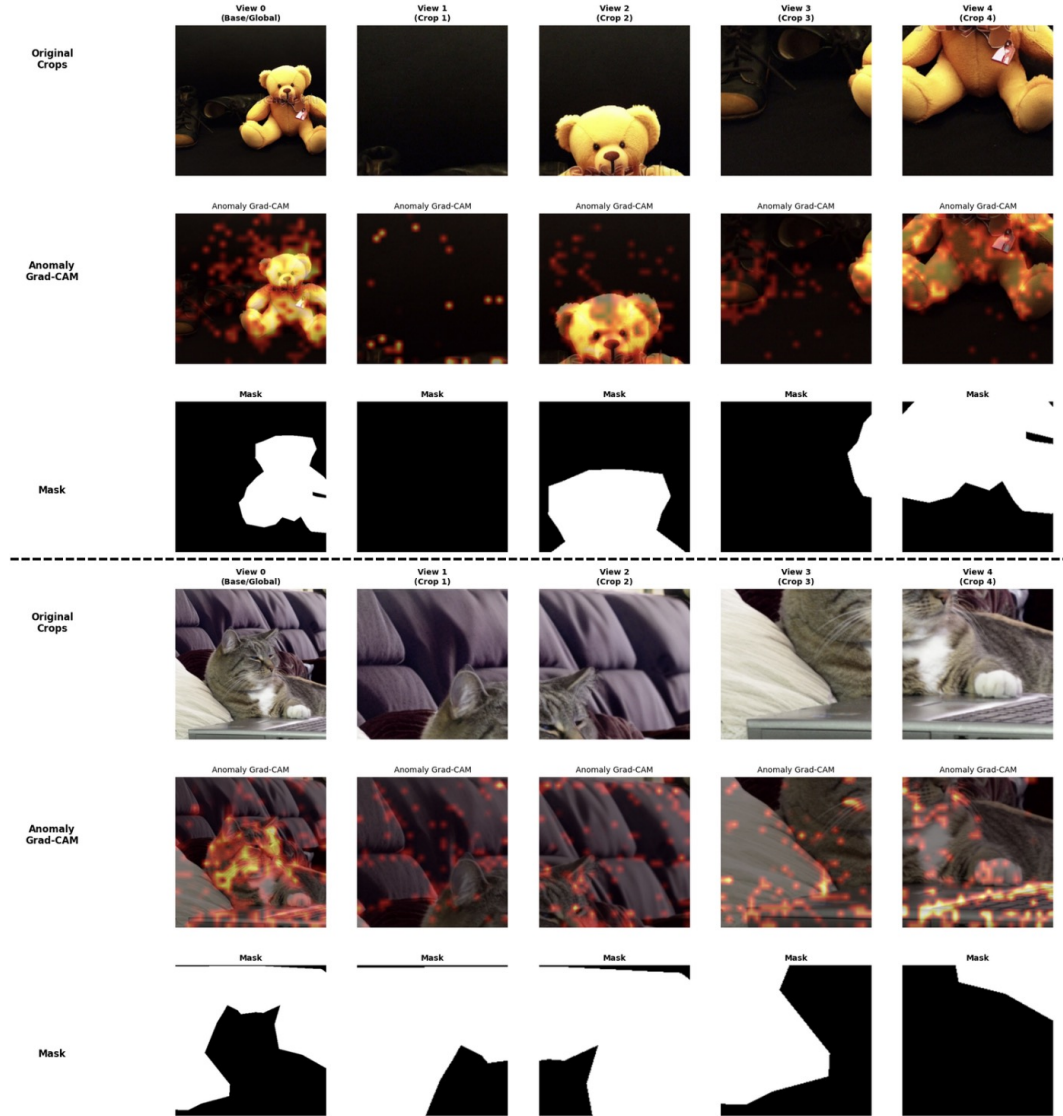


Figure 6. Anomaly OV zero shot test on 2 different images. Here we can see how ZS anomaly maps focus on image subjects. This behavior can lead to high localization performances when the map matches the inpainting mask (first image) and bad performances the rest of the times (second image)

| Metric    | AnomalyOV |      |      | R50-nodown |      |             | WaveRep |      |             |
|-----------|-----------|------|------|------------|------|-------------|---------|------|-------------|
|           | DF        | FGSM | PGD  | DF         | FGSM | PGD         | DF      | FGSM | PGD         |
| iou_topk  | 0.02      | 0.02 | 0.02 | 0.02       | 0.03 | 0.03        | 0.13    | 0.11 | <b>0.15</b> |
| mass_frac | 0.39      | 0.40 | 0.40 | 0.39       | 0.40 | 0.40        | 0.46    | 0.44 | <b>0.47</b> |
| pr_auc    | 0.42      | 0.42 | 0.42 | 0.48       | 0.56 | <b>0.59</b> | 0.57    | 0.53 | 0.58        |
| roc_auc   | 0.52      | 0.52 | 0.52 | 0.55       | 0.66 | <b>0.72</b> | 0.65    | 0.60 | 0.66        |

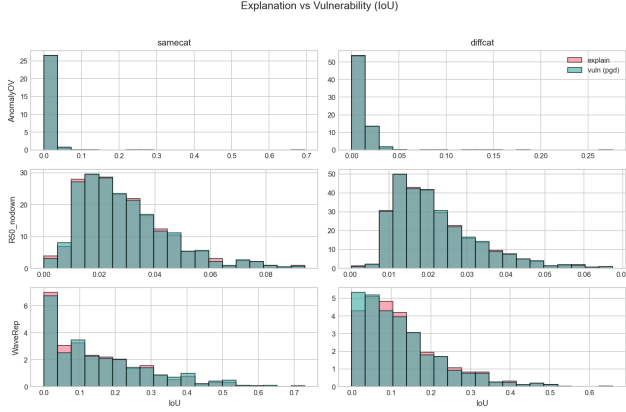
Table 1. Aggregated adversarial vulnerability localization across detectors (AnomalyOV, R50-nodown, WaveRep) and attacks (Deep-Fool = DF, FGSM, PGD).

| Metric    | Image Type | AnomalyOV | R50-nodown  | WaveRep     |
|-----------|------------|-----------|-------------|-------------|
| iou_topk  | diffcat    | 0.02      | 0.03        | <b>0.13</b> |
|           | samecat    | 0.02      | 0.03        | <b>0.15</b> |
| mass_frac | diffcat    | 0.49      | <b>0.72</b> | 0.56        |
|           | samecat    | 0.31      | <b>0.53</b> | 0.37        |
| pr_auc    | diffcat    | 0.51      | <b>0.68</b> | <b>0.68</b> |
|           | samecat    | 0.34      | 0.50        | <b>0.51</b> |
| roc_auc   | diffcat    | 0.51      | <b>0.72</b> | 0.69        |
|           | samecat    | 0.50      | <b>0.72</b> | 0.67        |

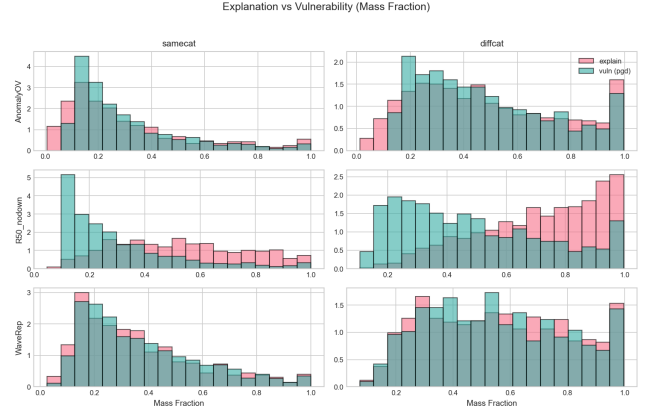
Table 2. Explainability performance on same-category vs different-category manipulations.

| Metric    | Image Type | AnomalyOV | R50-nodown  | WaveRep     |
|-----------|------------|-----------|-------------|-------------|
| iou_topk  | diffcat    | 0.02      | 0.03        | <b>0.12</b> |
|           | samecat    | 0.02      | 0.03        | <b>0.14</b> |
| mass_frac | diffcat    | 0.49      | 0.49        | <b>0.54</b> |
|           | samecat    | 0.30      | 0.31        | <b>0.36</b> |
| pr_auc    | diffcat    | 0.51      | <b>0.64</b> | 0.63        |
|           | samecat    | 0.33      | 0.45        | <b>0.49</b> |
| roc_auc   | diffcat    | 0.52      | <b>0.64</b> | 0.63        |
|           | samecat    | 0.52      | 0.64        | <b>0.65</b> |

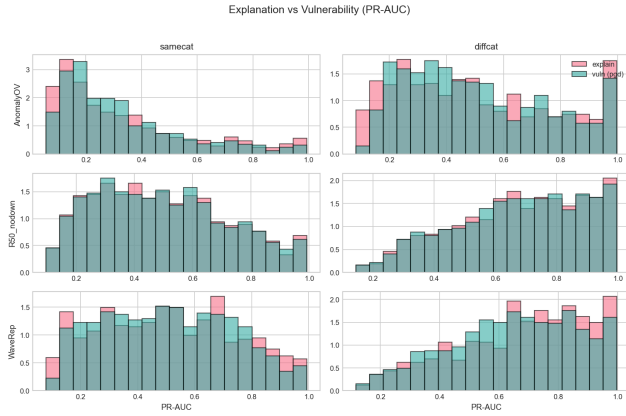
Table 3. Adversarial vulnerability localization results for same-category vs different-category manipulations.



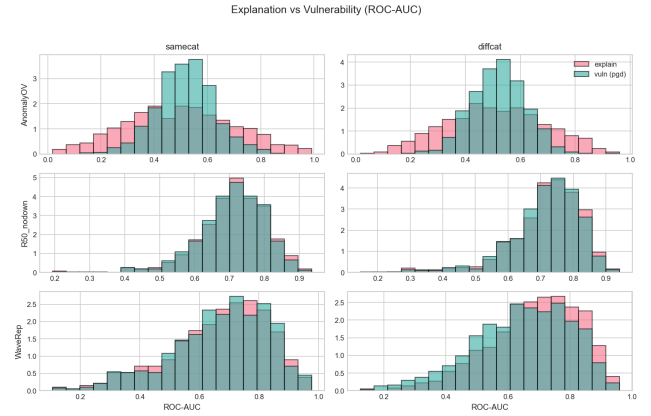
(a) IoU@TopK



(b) Mass-in-mask fraction



(c) PR-AUC



(d) ROC-AUC

Figure 7. Comparison between explainability maps and adversarial vulnerability maps across detectors and metrics.